



**SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING**  
(Deemed to be University)  
**Department of Mathematics and Computer Science**

**CIE - II SOLUTIONS**

*Data Preparation · Principles of Data Wrangling*

UDSC-402: Data Analysis and Visualization

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**I. State whether the following statements are True or False.  $[3 \times 1 = 3]$**

1. Data leakage can occur even when the rows of the test set are never directly used to fit the model.

*Answer: TRUE — normalising or standardising the full dataset before splitting leaks summary statistics (min, max, mean, standard deviation) of the test rows into training. This is indirect data leakage.*

2. In the Interquartile Range (IQR) method of outlier detection, the factor  $k = 1.5$  is used to identify “extreme outliers” or “far outs”.

*Answer: FALSE —  $k = 1.5$  identifies ordinary outliers. Extreme outliers (“far outs”) use  $k = 3$  or more.*

3. Schema-on-read means that data must adhere to structural and syntactic constraints in order to be ingested.

*Answer: FALSE — that is schema-on-write (traditional warehouses). Schema-on-read defers structure enforcement until the data is actually used (HDFS / Amazon S3 style).*

**II. Fill in the blanks.  $[3 \times 1 = 3]$**

4. A column in which every row contains the exact same value carries no information for modelling and is called a \_\_\_\_\_ predictor.

*Answer: zero-variance (a column with very few unique values is a near-zero-variance predictor)*

5. Iterative Imputation, in which each feature with missing values is modelled as a function of the other features and the process is repeated, is also known as \_\_\_\_\_.

*Answer: MICE — Multivariate Imputation by Chained Equations (also called Fully Conditional Specification, FCS)*

6. A sample that provides representation of every group or “stratum” present in the data — though it may bias overall trends by over-representing small groups — is called a \_\_\_\_\_ sample.

*Answer: stratified*

**III. Match the following data transforms with their correct definitions.  $[4 \times 1 = 4]$**

| Column A — Transform       | Column B — Definition                                          |
|----------------------------|----------------------------------------------------------------|
| i. Normalization Transform | a. Encode a categorical variable into binary (dummy) variables |
| ii. One Hot Transform      | b. Change the distribution of a variable to be more Gaussian   |

|                               |                                                     |
|-------------------------------|-----------------------------------------------------|
| iii. Discretization Transform | c. Scale a variable to the range 0 to 1             |
| iv. Power Transform           | d. Encode a numeric variable as an ordinal variable |

**Answer:**

|                   |                    |                     |                    |
|-------------------|--------------------|---------------------|--------------------|
| $i \rightarrow c$ | $ii \rightarrow a$ | $iii \rightarrow d$ | $iv \rightarrow b$ |
|-------------------|--------------------|---------------------|--------------------|

**Q7. List the FIVE core data preparation tasks in a predictive modelling project and give a one-line explanation of each. State which task is normally performed first and why. [5 Marks]**

**Answer:**

- Data Cleaning — identifying and correcting mistakes or errors in the data (outliers, duplicate rows, zero-variance columns, missing values).
- Feature Selection — identifying the subset of input variables most relevant to the target, so irrelevant or redundant inputs do not mislead the model.
- Data Transforms — changing the scale or the probability distribution of variables (normalisation, standardisation, power, quantile, ordinal, one-hot, discretisation).
- Feature Engineering — deriving new input variables from the available data (boolean flags, summary statistics, decomposing compound variables, polynomial transforms).
- Dimensionality Reduction — creating compact lower-dimensional projections of the data (PCA, SVD, LDA, autoencoders, t-SNE).
- Order: Data cleaning is performed first, because errors, duplicates and missing values would otherwise corrupt every statistic computed by the later steps.
- Guiding philosophy: data preparation should best expose the unknown underlying structure of the problem to the learning algorithm — it is a step of discovery, not a fixed recipe.

**Q8. Explain the FIVE characteristics used to describe data in the data-wrangling framework. Answer in points, with one example for each. [5 Marks]**

**Answer:**

- Structure — the format and encoding of records and fields. Ranges from rectangular (fixed rows  $\times$  columns), to jagged (inconsistent fields, supported by JSON/XML), to heterogeneous (mixed record types). Example: a retail file mixing customer records with transaction records.
- Granularity — what a single record represents, described as coarse vs fine. Example: one record = one sales transaction (fine) versus one record = total sales per region per week (coarse).
- Accuracy — the quality and correctness of the values themselves. Example: a UPC with missing digits, misspelled company names, or a 12-hour timestamp with no AM/PM indicator.
- Temporality — a record represents an entity at a point in time, so a correct record can go stale. Example: a sales record stays accurate about the original transaction but becomes inaccurate about what a customer owns once the item is returned.

- Scope — (a) breadth of attributes (wide = many fields, narrow = few) and (b) population coverage. Example: if under-18s were scrubbed for legal reasons, the dataset will show a higher average age than the true population.
- Why they matter: these five are the key aspects of representational consistency — they are the targets of transformation during wrangling, and in the production stage they define the boundary of permissible variation.

**Q9. A student builds a classifier as follows: he first scales the ENTIRE dataset using MinMaxScaler, then splits it 67:33 into train and test, fits the model and reports 98% accuracy. In deployment, the model performs far worse. [5 Marks]**

- Name the problem and explain why it arises here.
- State the correct sequence of steps for a train-test split.
- State how the same problem is avoided under k-fold cross-validation.

**Answer (summary points):**

- (a) The problem is data leakage — specifically indirect leakage. MinMaxScaler computes the global minimum and maximum over all rows, so summary statistics of the test rows are used while preparing the training data.
- The model therefore gains an unrealistic advantage and the 98% figure is an optimistic, incorrect estimate of true performance — which is why it collapses on genuinely unseen data.
- (b) Correct train-test procedure: (1) split into train and test sets FIRST; (2) fit the transform on the TRAINING set only; (3) apply the fitted transform to both train and test; (4) evaluate the model.
- (c) Under k-fold cross-validation the data preparation must be fitted inside each fold, using only that fold's training portion, then applied to the held-out portion.
- This is achieved in practice with a Pipeline — an ordered sequence of transform steps ending in the model — passed as a single unit to `cross_val_score()`. What is evaluated is therefore the pipeline, not merely the model.
- Golden rule: the entire modelling pipeline must be prepared on the training data only. The fitted transform must also be saved along with the final model so that future data is processed identically.

**Q10. While profiling the FEC Candidate Master File you find: (i) Column 14 (state of the candidate's mailing address) has 57 distinct values, including the value “ZZ” and some missing entries; and (ii) Column 4 (election year registered for) ranges from 1990 to 2064, although your project concerns only the 2016 presidential election. [5 Marks]**

- Identify which type of profiling detects each issue.
- Explain why 57 values is a problem for Column 14.
- State the recommended action for the out-of-scope election years and justify it.

**Answer (summary points):**

- (a) Issue (i) is caught by individual-value profiling of the syntactic kind — checking whether each literal value belongs to the set of permissible values, via a lookup against a reference list of valid state codes.

- Issue (ii) is caught by set-based profiling — examining the distribution of values of a single field across many records (here, a summary count per election year).
- (b) 57 is legitimate for the state of the office sought (50 states + 5 territories + DC + the non-state value “US” for presidential candidates), but a mailing address can never be “US”, so only 56 values are legitimately possible.
- The lookup exposes one record carrying the erroneous state code “ZZ”. The missing values also count as syntactically invalid, because the data dictionary does not indicate that missing values should appear.
- (c) The recommended action is NOT deletion — it is to insert additional metadata, flagging every record whose Column 4 value is not “2016” as invalid or out of scope.
- Justification: judging relevance to the 2016 presidential election is a custom (proprietary) metadata decision specific to this project. Records for 1990 or 2064 are valid FEC data and may be perfectly useful to a different analysis, so flagging preserves them while excluding them from the current scope.
- Practical extension: repeat the same procedure column by column, generating a series of Boolean indicator columns showing whether each record's values are permissible under the data dictionary.